

KINGSTON 1 NW, RI, USA

WMO: 725110

Lat: 41.4911N

Lon: 71.5413W

Elev: 35

StdP: 100.90

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 54796

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -14.8	(c) -12.0	(d) -22.3	(e) 0.5	(f) -13.0	(g) -19.4	(h) 0.7	(i) -10.0	(j) 9.1	(k) -0.3	(l) 7.9	(m) 0.4	(n) 1.1	(o) N/A	(p) 0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.3	(c) 30.1	(d) 23.0	(e) 28.5	(f) 22.0	(g) 27.1	(h) 21.3	(i) 24.4	(j) 28.2	(k) 23.6	(l) 26.8	(m) 22.9	(n) 25.7	(o) 2.4	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.2	18.1	26.4	22.6	17.4	25.5	21.9	16.7	24.7	73.7	28.4	70.6	26.8	67.9	25.7	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.9	(b) 5.8	(c) 4.9	(e) -19.1	(f) 32.7	(g) 3.5	(h) 1.6	(i) -21.6	(j) 33.9	(k) -23.7	(l) 34.8	(m) -25.6	(n) 35.7	(o) -28.1	(p) 36.9		
(4) 6.9	(5) 5.8	(4) 4.9	(4) -19.1	(4) 32.7	(4) 3.5	(4) 1.6	(4) -21.6	(4) 33.9	(4) -23.7	(4) 34.8	(4) -25.6	(4) 35.7	(4) -28.1	(4) 36.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	-1.8	-1.1	2.7	8.3	13.6	18.2	21.8	21.0	17.4	11.5	6.0	1.4
	DBStd	9.20	5.62	4.96	4.70	3.94	3.78	3.31	2.70	2.84	3.43	4.32	4.58	4.94
	HDD10.0	1435	368	310	231	79	8	0	0	0	0	34	136	270
	HDD18.3	3331	626	543	486	302	154	43	4	8	57	216	369	526
	CDD10.0	1427	1	1	3	27	120	245	366	342	223	80	17	3
	CDD18.3	282	0	0	0	1	8	38	112	91	30	3	0	0
	CDH23.3	1841	0	0	0	15	71	266	743	602	132	12	0	0
	CDH26.7	394	0	0	0	3	15	58	181	121	15	1	0	0
Wind	WSAvg	2.0	2.0	2.0	2.4	2.4	2.2	1.9	1.7	1.5	1.8	2.0	1.9	1.9
Precipitation	PrecAvg	1300	110	97	125	118	107	93	79	105	109	105	127	121
	PrecMax	1783	315	211	312	329	244	364	187	323	276	379	330	286
	PrecMin	824	23	23	18	31	18	1	19	14	23	16	19	23
	PrecStd	200	58	43	62	62	55	73	47	68	54	62	64	61
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.0	13.7	18.9	25.4	28.7	31.1	32.6	31.5	28.8	25.2	19.2	16.1
		MCWB	12.5	10.8	12.5	16.7	20.2	22.9	24.8	24.3	23.1	18.6	15.6	14.2
	2%	DB	11.6	11.2	14.4	20.3	25.1	28.2	30.3	29.5	26.2	21.8	17.3	13.8
		MCWB	10.0	8.8	10.2	13.6	18.5	21.0	23.3	22.8	20.7	19.0	14.9	12.6
	5%	DB	9.1	9.1	12.0	17.3	22.8	26.3	28.7	27.9	24.8	20.3	15.7	12.1
		MCWB	7.6	6.8	8.9	12.2	17.1	19.8	22.5	21.8	20.5	17.5	13.1	10.8
	10%	DB	6.6	6.8	10.1	15.1	20.3	24.5	27.2	26.6	23.5	18.8	14.3	9.9
		MCWB	4.6	4.3	7.1	11.0	15.4	18.5	21.7	21.3	20.0	16.3	11.6	8.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.3	12.6	15.1	17.2	21.9	23.9	26.1	25.5	24.1	21.2	17.0	15.2
		MCDB	13.8	13.7	18.7	21.6	27.6	28.8	31.3	29.8	27.6	22.9	18.4	15.6
	2%	WB	10.6	10.2	11.3	15.0	20.2	22.4	24.7	24.2	23.0	19.9	15.7	13.3
		MCDB	11.3	11.5	13.2	16.7	24.2	26.7	29.1	27.2	25.6	21.5	16.8	14.0
	5%	WB	7.7	7.9	9.8	13.5	18.6	21.3	23.8	23.5	22.0	18.5	13.9	11.6
		MCDB	9.0	9.6	12.1	15.7	21.7	24.6	27.4	26.0	23.8	20.2	15.2	12.5
	10%	WB	5.1	5.3	7.4	11.9	17.2	20.3	23.0	22.7	21.2	17.2	11.7	9.3
		MCDB	6.5	7.3	9.5	14.1	20.0	23.1	26.1	25.0	22.9	19.0	13.6	10.7
Mean Daily Temperature Range	5% DB	MDBR	9.0	9.6	9.5	10.2	10.0	9.6	9.3	9.7	10.1	10.2	10.0	8.8
		MCDBR	11.6	11.7	12.1	13.7	13.1	12.0	10.7	10.5	10.7	10.6	10.7	10.3
		MCWBR	11.2	9.3	8.0	7.0	6.9	6.0	5.0	5.3	6.3	7.3	9.4	9.9
	5% WB	MCDBR	12.3	12.0	10.7	9.5	10.8	10.2	9.0	8.5	8.1	8.8	10.1	10.1
		MCWBR	11.9	10.7	8.4	7.1	6.4	5.7	4.6	5.0	5.4	7.2	9.9	10.1
Clear-Sky Solar Irradiance	taub	0.305	0.313	0.336	0.374	0.418	0.461	0.485	0.445	0.376	0.343	0.315	0.303	
	taud	2.500	2.477	2.454	2.381	2.288	2.208	2.149	2.254	2.421	2.520	2.557	2.556	
	Ebn at Noon	853	900	914	897	860	821	796	818	859	854	836	828	
	Edh at Noon	74	87	100	116	130	142	149	130	103	83	70	65	
All-Sky Solar Radiation	RadAvg	1.89	2.77	3.78	4.88	5.39	5.92	6.06	5.44	4.44	3.00	2.05	1.58	
	RadStd	0.15	0.27	0.27	0.52	0.52	0.59	0.32	0.29	0.33	0.28	0.15	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.61	N/A	N/A	+0.80	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+96	+47

Nomenclature: See separate page